Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

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BATCH:A1

DATE OF SUBMISSION:17/04/24

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Kavya Gada

Java Exp:-10

* Aim:- To implement dynamic polymorphism, final keyword and garbage.

→ Description

1) Final Keyword - The final keyword in java is used to restrict the user. The java keyword can be used in many context. Variables, method and class can be final. A final variable that have no value it is called blank final variable or unutilized final variable. It can be initialized in the constructor only.

2) Final variables, method and class.

i) → If you make a variable final, you cannot change the value of the final variable.

Example :-

```
Class Bikeq {
  Final int speedlimit =90;
  void run() {
    speed limit =400; }

  public static void main (String args[]) {
    Bikeq b = new Bikeq();
    b.run();
  }
```

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ii) Final Method

→ If you make any method as finally you cannot override it.

Example:-

```
class Bike {
 final void run ();
 }
 class Honda extends Bike {
 void run() {
 System.out.println ("without look mph"); }

 public static void main (String args []) {
  Honda h = new Honda ();
   h.run();
  }
}
```

iii) Final Class

→ If you make a class finally you cannot extend it.

Example:-

```
final class Bike {
class Honda extends Bike {
 void run() {
  System.out.println ("Running"); }

 public static void main (String args []) {
  Honda h = new Honda ();
  h.run ();
 }
}
```

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③ Overriding and Dynamic Method Dispatch

→ If a sub-class has the same name method as declared in the parent class, it is known as Method Overriding
If a subclass provides the specific implementation of the method that has been declared by one of its parent class, it is known as Method overriding.
Runtime polymorphism and Dyamic Method dispatch is a process in which a call to an overridden method is resolved at runtime rather than compile time. In this process, an overridden method is called through the reference variable of a superclass. The determination of the method to be called is based on the object being to by the reference variable.

4) Garbage collection (finalize()).
→ Garbage collection is a process of reclaiming the runtime unused memory automatically. In other words, It is a way to destroy the unused objects The finalize() method is invoked each time before the object is garbage collected. This method can be used to perform cleanup processing. The method is defined in Object class as:-

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10)c) WAP to create an object of a class, delete the same object by calling System.gc() and display a message that the "Object has been deleted."

Program:-

```
Class ObjectDelete {
public static void main (String args[]) {
    MyClass o = new MyClass();
    System.out.println("Object Created");
    o = null;
    System.gc();
    System.out.printh("Garbage collection is requested");
  }
}
class MyClass {
  protected void finalize() Throws Throwable
    {
      System.out.println("Object has been deleted");
    }
}
```

Output:-

```
D:java > javac ObjectDelete.java
D:java > java ObjectDelete
D:java > Object Created
         Garbage Collection is Request
         Object has been deleted.
```


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EXPERIMENT NO.10

Q1) Demonstrate using a suitable example that a base class reference variable can point to a child class object or a base class object using the concept of dynamic method dispatch (dynamic polymorphism).

CODE :

```
class Shape {
void draw() {
System.out.println("Drawing a shape");
}
}
class Circle extends Shape {
void draw() {
System.out.println("Drawing a circle");
}
}
class Rectangle extends Shape {
void draw() {
System.out.println("Drawing a rectangle");
```


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```
}
}
public class Q10a {
public static void main(String[] args) {
Shape shape1 = new Circle();
shape1.draw();
Shape shape2 = new Rectangle();
shape2.draw();
Shape shape3 = new Shape();
shape3.draw();
}
}
```

OUTPUT :

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```
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>cd C:\java program

C:\java program>javac Q10a.java

C:\java program>java Q10a
Drawing a circle
Drawing a rectangle
Drawing a shape

C:\java program>
```


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Q2) Adwita , 8th Grade student wants to write a functions to calculate simple interest, compound interest. She wants to keep same (final) rate of interest for every input of principal and time. She wants to ensure that the declared functions are not overridden in any subclasses and the class is not inherited by any other class. Help her to declare the variables methods and classes and write the code for the same using final keyword.

CODE :

```
final class InterestCalculator {
private final double rate;
public InterestCalculator(double rate) {
this.rate = rate;
}
public final double calculateSimpleInterest(double principal, double time) {
return (principal * rate * time) / 100;
}
```


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```
public final double calculateCompoundInterest(double principal, double
time) {
return principal * Math.pow(1 + rate / 100, time) - principal;
}}
public class Q10b {
public static void main(String[] args) {
InterestCalculator calculator = new InterestCalculator(5);
double principal = 1000;
double time = 2;
double simpleInterest = calculator.calculateSimpleInterest(principal, time);
System.out.println("Simple Interest: " + simpleInterest);
double compoundInterest =
calculator.calculateCompoundInterest(principal, time);
System.out.println("Compound Interest: " + compoundInterest);
}
}
```


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OUTPUT :

```
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>cd C:\java program

C:\java program>javac Q10b.java

C:\java program>java Q10b
Simple Interest: 100.0
Compound Interest: 102.5

C:\java program>
```


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Q3) WAP to create an object of a class, delete the same object by calling System. gc () and display a message that the "object has been deleted".

CODE :

```
class MyClass implements AutoCloseable {
MyClass() {
System.out.println("Object created");
}
public void close() {
System.out.println("Object has been deleted");
}
}
public class Main {
public static void main(String[] args) {
try (MyClass obj = new MyClass()) {
} catch (Exception e) {
e.printStackTrace();
```


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```
}
}
}
```

OUTPUT :

```
Microsoft Windows [Version 10.0.19045.4291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>cd C:\java program

C:\java program>javac Main.java

C:\java program>java Main
Object created
Object has been deleted

C:\java program>_
```


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CONCLUSION:

•Dynamic polymorphism in Java enables the invocation of overridden methods at runtime based on the actual type of the object.

•The final keyword is used to prevent further modification, such as subclassing a class, overriding a method, or changing the value of a variable.

• Garbage collection automatically reclaims memory occupied by unreachable objects, enhancing memory management and preventing memory leaks in Java applications.

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